

Modeling Rate of Change and Sensitivity

Group Activity: The Sliding Ladder

Instructor Solution

A 10-ft ladder is leaning against a vertical wall. Let

x = distance of the bottom of the ladder from the wall,

and

y = height of the top of the ladder above the ground.

Task 1: Build the Model

- The wall, ground, and ladder form a right triangle. The horizontal side is x , the vertical side is y , and the hypotenuse is 10 ft.
- The quantities x and y both change as the ladder moves.
- The length of the ladder remains constant:

$$L = 10 \text{ ft.}$$

- By the Pythagorean theorem,

$$\boxed{x^2 + y^2 = 100}.$$

- Although x and y change, the ladder remains 10 ft long. Therefore, the two changing quantities must always satisfy

$$x^2 + y^2 = 100.$$

This equation represents the **constraint** governing the physical system.

Task 2: Predict Before Calculating

Suppose the bottom of the ladder moves away from the wall.

- The quantity x is **increasing**.
- The quantity y is **decreasing**.

- c. Since an increase in x produces a decrease in y , we expect

$$\boxed{\frac{dy}{dx} < 0}.$$

The negative sign represents an inverse relationship between the two changing quantities: as the bottom moves farther from the wall, the top moves downward.

Task 3: How Sensitive is y to x ?

The constraint equation is

$$x^2 + y^2 = 100.$$

- a. Differentiate implicitly with respect to x :

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(100).$$

Since y depends on x ,

$$2x + 2y \frac{dy}{dx} = 0.$$

- b. Solving for $\frac{dy}{dx}$,

$$2y \frac{dy}{dx} = -2x,$$

so

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}}.$$

This derivative measures the **sensitivity of the vertical height y to changes in the horizontal position x** .

- c. When $x = 6$ ft,

$$6^2 + y^2 = 100.$$

Thus,

$$y^2 = 64,$$

and since height is positive,

$$\boxed{y = 8 \text{ ft}}.$$

d. Therefore,

$$\frac{dy}{dx} = -\frac{6}{8} = -\frac{3}{4}.$$

Hence,

$$\boxed{\frac{dy}{dx} = -0.75}.$$

e. The negative sign tells us that an increase in x produces a decrease in y .

Thus,

$$x \uparrow \implies y \downarrow.$$

f. The magnitude 0.75 describes the local sensitivity of the height to horizontal position.

Near $x = 6$ ft, a small change Δx produces approximately

$$\Delta y \approx \frac{dy}{dx} \Delta x.$$

Therefore,

$$\boxed{\Delta y \approx -0.75 \Delta x}.$$

For example, if

$$\Delta x = 0.1 \text{ ft},$$

then

$$\Delta y \approx (-0.75)(0.1) = -0.075 \text{ ft}.$$

Thus, near this position, moving the bottom 0.1 ft farther from the wall causes the top to move downward by approximately 0.075 ft.

Task 4: Bring Time into the Model

Suppose the bottom of the ladder is being pulled away from the wall at

$$\frac{dx}{dt} = 2 \text{ ft/s}.$$

a. Since the top of the ladder moves downward as the bottom moves away from the wall, we predict

$$\boxed{\frac{dy}{dt} < 0}.$$

b. Starting with

$$x^2 + y^2 = 100,$$

differentiate both sides with respect to time:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0.$$

Therefore,

$$y \frac{dy}{dt} = -x \frac{dx}{dt},$$

and hence

$$\boxed{\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}}.$$

c. When $x = 6$ ft, we know that

$$y = 8 \text{ ft.}$$

Therefore,

$$\frac{dy}{dt} = -\frac{6}{8}(2).$$

Thus,

$$\boxed{\frac{dy}{dt} = -1.5 \text{ ft/s}}.$$

d. The negative sign indicates that the height is decreasing.

Therefore, when the bottom of the ladder is 6 ft from the wall, the top of the ladder is moving **downward at 1.5 ft/s**.

Task 5: Make the Connection

At $x = 6$ ft,

$$\frac{dy}{dx} = -0.75, \quad \frac{dx}{dt} = 2 \text{ ft/s},$$

and

$$\frac{dy}{dt} = -1.5 \text{ ft/s}.$$

- a. These quantities satisfy

$$\boxed{\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}}.$$

Indeed,

$$(-0.75)(2) = -1.5.$$

- b. Therefore,

Sensitivity of y to x $\times$ Rate at which x changes $=$ Rate at which y changes
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- c. In words:

How strongly y responds to a change in x , multiplied by how quickly x itself is changing, determines how quickly y changes with time.

This connects **implicit differentiation**, **sensitivity**, and **related rates**.

Task 6: What Happens if We Change the Position?

We keep the pulling rate fixed:

$$\frac{dx}{dt} = 2 \text{ ft/s.}$$

From the constraint,

$$x^2 + y^2 = 100,$$

we have

$$y = \sqrt{100 - x^2}.$$

Also,

$$\frac{dy}{dx} = -\frac{x}{y},$$

and

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = -\frac{2x}{y}.$$

The results for the four groups are shown below.

x (ft)	y (ft)	$\frac{dy}{dx}$	$\frac{dy}{dt}$ (ft/s)
2	$4\sqrt{6} \approx 9.80$	-0.204	-0.408
6	8	-0.750	-1.500
8	6	-1.333	-2.667
9	$\sqrt{19} \approx 4.36$	-2.065	-4.129

Final Group Challenge

- a. Among the four positions considered, the height is most sensitive when

$$\boxed{x = 9 \text{ ft}}.$$

At this position,

$$\left| \frac{dy}{dx} \right| \approx 2.065,$$

which is the largest sensitivity among the four positions.

- b. Since

$$\left| \frac{dy}{dx} \right| = \frac{x}{y},$$

as the ladder becomes more horizontal,

$$x \rightarrow 10, \quad y \rightarrow 0.$$

Therefore,

$$\boxed{\left| \frac{dy}{dx} \right| \rightarrow \infty}.$$

Thus, the vertical position of the top becomes increasingly sensitive to horizontal movement of the bottom.

- c. When the ladder is nearly vertical, a small horizontal movement of its bottom produces relatively little vertical movement of its top.

When the ladder is nearly horizontal, the same small horizontal movement can produce a much larger vertical movement.

Therefore, the **sensitivity of the system depends on its current configuration.**

d. The quantities

$$\frac{dy}{dx} \quad \text{and} \quad \frac{dy}{dt}$$

describe two different aspects of the system.

The derivative

$$\frac{dy}{dx}$$

measures how sensitive the height y is to a change in the horizontal position x .

The derivative

$$\frac{dy}{dt}$$

measures how quickly the height is actually changing with respect to time.

They are connected through

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}.$$

Main Modeling Idea

Sensitivity and Related Rates

$$\underbrace{\frac{dy}{dx}}_{\text{How sensitive is } y \text{ to } x?} \times \underbrace{\frac{dx}{dt}}_{\text{How fast is } x \text{ changing?}} = \underbrace{\frac{dy}{dt}}_{\text{How fast is } y \text{ changing?}}$$

When quantities in a physical system are connected, a change in one quantity can force another quantity to change. Derivatives allow us to model both the **sensitivity** between quantities and their **rates of change with time**.